B5: E-Graph-Augmented Tool-in-the-Loop RTL Verification

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P2: Prototype & Pilot Results | Due November 9, 2026, 11:59 PM ET

I. Project Goal and Updates

Our project asks whether E-graph semantic feedback helps an LLM verify RTL more effectively and efficiently than raw execution feedback. The proposal compares text-only (A), raw-coverage (B), and E-graph-enhanced (C) stimulus generation. That track already has exploratory coverage pilots. Our separately authorized *test2* extension fixes the stimuli and asks the LLM to generate passive SystemVerilog monitors. It measures valid fault detection, false alarms, and cost. This separates the quality of a generated checker from the model's ability to choose inputs and provides a complementary test of the proposal's central semantic-feedback hypothesis.

This endpoint change is explicit: changing a monitor under fixed stimuli does not improve DUT coverage. Raw source coverage supplies feedback and workload context; monitor quality is the outcome. We do not claim an official CVDP or Pass@1 evaluation. Current results are descriptive pilots, and a negative C/B comparison remains useful evidence. Our next priority is a bounded comparison on a different protocol IP rather than extending a general verification-tool qualification framework.

II. Method Overview

The implemented loop freezes the DUT, specification, inputs, reference, model settings, and resource caps. Python orchestrates GPT candidate generation, compilation and public development feedback, per-round snapshots, final release, hidden grading, replay, and reporting. A Rust `egg` pipeline forms restricted equivalence groups. Typed complete guards and independent Z3 checks retain widths, signedness, activation conditions, and clock/reset context. These checks establish limited two-state combinational relations, not general sequential or four-state equivalence. Equivalent guards retain their distinct original source counters.

A Claude-authored specification reference remains independent of GPT's candidate monitors. Candidates cannot grade themselves. Public prompts exclude the reference implementation, hidden faults, final evaluation traces, and solver witnesses. B receives the specification, permitted RTL, fixed development observations, raw counters, and its own development feedback. C receives the same information plus validated E-graph relations. Within each arm, sessions are independent; all rounds are frozen before hidden scoring. The final candidate is selected by the registered validity/compilation rule, not its hidden score.

The controller study uses `gpt-6.1-sol` with low reasoning, three sessions per arm and six rounds per session, for 36 generation calls. Each call has the same 64k-input/8,192-output token caps. Equal caps do not imply equal spending. A final-post `trace_done` marker exposes the finite evaluation boundary to a passive monitor without adding a DUT input or clock edge.

Scoring first checks finite correct-DUT controls. Fault credit requires an independent reference-confirmed observable violation and an eligible candidate alarm; earlier alarms cannot earn credit. Invalid candidates, unknowns, unobservable fault types, and failures remain in the evidence. Repeated alarms and two simulators do not create additional defect types. API tokens, compilation, physical DUT simulation/cycles, monitor replay, and wall time are recorded separately. Without a token-matched placebo, B/C compares the complete feedback package rather than isolating semantic content from extra context.

III. Prototype and Pilot Results

What runs end to end. The saved multi-cycle request/response controller study includes fixed physical observations, monitor generation, compilation and development feedback, candidate freeze, hidden grading, saved-response replay, and a result report. Concrete inputs are `test2/configs/controller_done_study_v1.json` and `test2/tasks/axis_length_done_v1/` input/evaluation freezes. The generation entry is `test2/scripts/controller_done_generate_v2.py`; the postprocessing entry is `test2/scripts/controller_done_live_post_v2.py`. The live run's `release.json` and `final-candidates.json` bind its candidates. The live and saved-response replay paths have separately saved scores.

The original final audit adapter failed on a catalog-field mismatch. That failure is preserved. A new read-only adapter recovered the audit from already saved evidence without rerunning model generation or physical/formal experiments. Thus the prototype has executed end to end with documented recovery; we do not present it as a fresh, clean single-command run. Historical commands, exits, and recovery are retained in the postprocessing progress files.

Observed comparison. All six final selected monitors pass 18 finite correct-control cases each and detect 11/11 confirmed

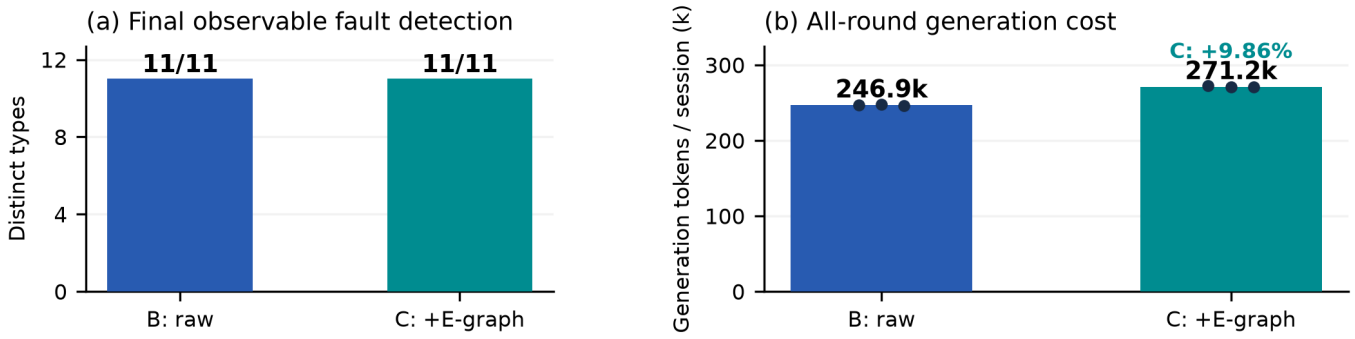


Fig. 1. Saved controller fixed-stimulus pilot: three independent sessions per arm, six rounds each (36 generation calls). All six final monitors detect 11/11 confirmed observable types; the raw denominator remains 12, including N10 whose observability was not confirmed. Right: mean actual generation input+output tokens over all rounds; dots are session totals. B totals 740,643 tokens and C 813,649, a 9.8571% C increase. Cache/reasoning subcounts are included, not added again. This figure excludes compilation/simulation/wall costs and is not an invoice, significance test, or general correctness claim.

observable injected fault types (Fig. 1). The original denominator is 12 types/144 fault pairs; N10 is retained as not confirmed observable. B uses 740,643 generation tokens across its 18 rounds, versus 813,649 for C: C uses 9.8571% more, with no extra final fixed-trace kills. Per-session means are 246,881 and 271,216.33 tokens. The score is a finite, unattributed case/type union, not a proof that each monitor slot checks a particular semantic category.

Unknowns are not converted to passes: 38 reference-unknown pairs remain per candidate, and one B candidate has 40 total unresolved pairs after its candidate unknowns are included. Earlier invalid rounds and their generation costs remain. The separate formal schedule retains 75 PASS, 104 FAIL, and 361 TIMEOUT statuses over 540 jobs, including 74 unreproduced/unresolved BMC witnesses. Native failures are not pooled as confirmed detected faults, and no 128-cycle completion proof follows.

The exact numeric source is `test2/results/controller-done-live-acceptance-v3/audit.json`; the scoped report is `test2/results/controller-done-live-report-v4/report.md`. This P2 figure was rendered from that saved cohort; no research experiment was rerun to prepare this report. The observed endpoint is saturated, so the result supports no additional final C benefit on this workload, not a universal claim that E-graph feedback cannot help verification.

IV. Next Steps

Main risk. A small, saturated fault set offers little room to distinguish B from C, while longer C prompts confound semantic value with context cost. One inspected IP and three sessions per arm do not establish cross-IP performance or statistical significance. We will preserve this tie and its costs rather than change labels or targets to obtain a positive result.

Toward Milestone 3 on Nov 23, our next bounded pilot is OpenTitan I2C READB with one ACKing target. Existing assets include original RTL/specification, a fixed driver, calibration observations, a finite Claude reference, and validated guard relations. A one-page task card, a 32-field public observation mapping, role allowlists, and a reference-scope review prompt are prepared. The new test2 SV adapter, disjoint hidden inputs, and physically observable DUT mutants are *not implemented or tested*. Old trace-corruption controls are not RTL defects and will not substitute for physical mutant qualification.

Before candidate calls, we will check only the adopted input/reset/first-sample and pre/post path, positive/negative/unknown controls, independent reference scope, fixed data, and prompt isolation. A fresh independent Claude review must confirm the existing finite oracle's permitted scope; any required oracle change needs a separate Claude author and fresh reviewer. GPT will not silently repair it. Pending/invalid traces and the existing stretch, liveness, extra-STOP, and 256-byte completion limits remain. Generic formal-tool qualification is paused with its failures and costs preserved; it is not an unlimited prerequisite for this separate finite simulation study.

The proposed small I2C comparison has two independent sessions per arm and three rounds, with identical model/caps, selection, fixed stimuli, and hidden grading. Actual settings, rates, and service budget still require a concrete online decision. No new I2C live results or paid calls are claimed here.

Midterm feedback response pending. Actual instructor/TA technical feedback has not been provided or located in the bounded local search. We will record the original comment, resulting change, evidence, and limitation when it is available. The user's prioritization of cross-IP work and internal reviews are not substituted for classroom feedback.